

Co<sup>+</sup> ionisation limit 63564.60 cm<sup>-1</sup> = 7.88101 eV (157.320 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle 0 || er || j \rangle$  in  $e a_0$   
 Matrix elements are measured or NIST values; \* = high-accuracy theory, = model calculation.  
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



| Transition                  | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ \langle J U J\rangle $ (e.a.) | $A$ (s <sup>-1</sup> ) | $\Gamma/2\pi$ ur. (MHz) | branch (%) | Use |
|-----------------------------|--------------------|--------------------|-------------|---------------------------------|------------------------|-------------------------|------------|-----|
| $a^1F_{3/2} - w^1F_{3/2}^o$ | 233.9379           | 233.8662           | 1281.5042   | 1.525                           | 9.200e+07              | 36.17                   | 40.5       |     |
| $a^1F_{5/2} - x^1G_{5/2}^o$ | 240.7983           | 240.7251           | 1244.9940   | 6.391                           | 4.940e+08              | 75.58                   | 98.9       |     |
| $a^1F_{5/2} - x^1G_{3/2}^o$ | 241.5191           | 241.4457           | 1241.2787   | 4.856                           | 4.240e+08              | 75.79                   | 98.8       |     |
| $a^1F_{3/2} - x^1G_{3/2}^o$ | 241.6024           | 241.5290           | 1240.8506   | 3.999                           | 3.830e+08              | 69.2                    | 82.8       |     |
| $a^1F_{3/2} - x^1F_{5/2}^o$ | 242.5668           | 242.4931           | 1235.9172   | 5.112                           | 3.710e+08              | 63.66                   | 96.1       |     |
| $a^1F_{3/2} - x^1F_{3/2}^o$ | 243.2947           | 243.2209           | 1232.3193   | 4.259                           | 3.190e+08              | 58.95                   | 86.1       |     |
| $a^1F_{3/2} - x^1F_{5/2}^o$ | 243.7396           | 243.6657           | 1232.9702   | 3.313                           | 2.560e+08              | 56.84                   | 71.7       |     |
| $a^1F_{3/2} - x^1F_{3/2}^o$ | 243.9778           | 243.9038           | 1228.7695   | 2.838                           | 2.810e+08              | 54.88                   | 81.5       |     |
| $b^1F_{3/2} - w^1F_{3/2}^o$ | 251.1774           | 251.1018           | 1193.5487   | 2.653                           | 9.000e+07              | 28.42                   | 50.4       |     |
| $a^1F_{3/2} - x^1D_{3/2}^o$ | 252.2122           | 252.1363           | 1188.6519   | 4.242                           | 2.840e+08              | 51.34                   | 88         |     |
| $a^1F_{3/2} - x^1D_{5/2}^o$ | 252.9727           | 252.8967           | 1185.0781   | 3.537                           | 2.610e+08              | 49.74                   | 83.5       |     |
| $a^1F_{3/2} - x^1D_{3/2}^o$ | 253.6724           | 253.5961           | 1181.8097   | 2.723                           | 2.300e+08              | 46.81                   | 78.2       |     |
| $a^1F_{3/2} - x^1D_{5/2}^o$ | 254.5017           | 254.4253           | 1177.9586   | 2.150                           | 2.840e+08              | 46.81                   | 76.6       |     |
| $b^1F_{3/2} - y^1F_{3/2}^o$ | 340.6091           | 340.5114           | 880.1658    | 4.674                           | 1.120e+08              | 23.07                   | 97.3       |     |
| $b^1F_{3/2} - y^1G_{5/2}^o$ | 345.4498           | 345.3508           | 867.8323    | 5.677                           | 1.320e+08              | 21.51                   | 97.7       |     |
| $b^1F_{3/2} - y^1D_{5/2}^o$ | 350.7315           | 350.6312           | 854.7635    | 3.484                           | 9.500e+07              | 18.29                   | 82.7       |     |

$\lambda, \nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

### <sup>59</sup>Co hyperfine structure ( $I = 7/2$ )

| Level                           | A (MHz)      | B (MHz)    | Shift of each F level from the centre of gravity (mHz) |                |                |                |               |               |                |                |
|---------------------------------|--------------|------------|--------------------------------------------------------|----------------|----------------|----------------|---------------|---------------|----------------|----------------|
| a <sup>1</sup> F <sub>5/2</sub> | 450.2835(9)  | 139.22(3)  | F=1: -8604.15                                          | F=2: -7723.47  | F=3: -6398.31  | F=4: -4623.69  | F=5: -2392.99 | F=6: +302.08  | F=7: +3471.47  | F=8: +7126.77  |
| a <sup>3</sup> F <sub>7/2</sub> | 490.5672(31) | 94.50(3)   | F=0: -7675.82                                          | F=1: -7194.90  | F=2: -6231.12  | F=3: -4780.62  | F=4: -2837.64 | F=5: -394.44  | F=6: +2558.61  | F=7: +4603.08  |
| a <sup>3</sup> F <sub>5/2</sub> | 613.348(3)   | 67.54(6)   | F=1: -6863.99                                          | F=2: -5956.59  | F=3: -3838.25  | F=4: -1400.30  | F=5: +1668.86 | F=6: +5383.68 |                |                |
| a <sup>3</sup> F <sub>3/2</sub> | 1042.9828(9) | 67.618(18) | F=2: -7003.91                                          | F=3: -3923.26  | F=4: +229.35   | F=5: +5492.56  |               |               |                |                |
| b <sup>1</sup> F <sub>5/2</sub> | 828.800(3)   | -118.7(3)  | F=1: -16008.82                                         | F=2: -14334.26 | F=3: -11825.95 | F=4: -8488.14  | F=5: -4326.47 | F=6: +651.99  | F=7: +6438.75  | F=8: +13023.92 |
| b <sup>3</sup> F <sub>7/2</sub> | 668.918(3)   | -79.23(21) | F=0: -10577.91                                         | F=1: -9900.90  | F=2: -8548.51  | F=3: -6523.97  | F=4: -3832.13 | F=5: -479.45  | F=6: +3525.97  | F=7: +8174.44  |
| b <sup>3</sup> F <sub>5/2</sub> | 810.0(24)    | -60.00     | F=1: -15620.77                                         | F=2: -13992.12 | F=3: -11550.93 | F=4: -8299.34  | F=5: -4240.21 | F=6: +622.89  | F=7: +6285.67  | F=8: +12743.13 |
| z <sup>1</sup> F <sub>5/2</sub> | 655.4(24)    | 30.00      | F=0: -10305.69                                         | F=1: -9653.40  | F=2: -8348.21  | F=3: -6388.90  | F=4: -3773.62 | F=5: -499.93  | F=6: +3435.23  | F=7: +8035.54  |
| z <sup>3</sup> F <sub>7/2</sub> | 772.0(21)    | 210.00     | F=2: -17475.45                                         | F=3: -15193.91 | F=4: -12144.21 | F=5: -8319.81  | F=6: -3712.54 | F=7: +1687.43 | F=8: +7891.56  | F=9: +14912.92 |
| z <sup>3</sup> F <sub>5/2</sub> | 559.1(21)    | 30.00      | F=1: -6273.92                                          | F=2: -5164.27  | F=3: -3496.58  | F=4: -1267.00  | F=5: +1529.62 | F=6: +4899.71 |                |                |
| z <sup>3</sup> G <sub>5/2</sub> | 517.1(27)    | 180.00     | F=1: -9872.44                                          | F=2: -8863.88  | F=3: -7345.67  | F=4: -5311.40  | F=5: -2752.49 | F=6: +341.78  | F=7: +3984.26  | F=8: +8189.95  |
| z <sup>3</sup> D <sub>5/2</sub> | 751.0(27)    |            | F=0: -11827.93                                         | F=1: -11076.95 | F=2: -9575.00  | F=3: -7322.06  | F=4: -4318.14 | F=5: -562.24  | F=6: +3942.64  | F=7: +9199.50  |
| z <sup>3</sup> G <sub>3/2</sub> | 648.2(15)    | 150.00     | F=0: -6978.64                                          | F=1: -6545.75  | F=2: -5676.92  | F=3: -4366.03  | F=4: -2603.88 | F=5: -378.23  | F=6: +2326.21  | F=7: +5527.83  |
| z <sup>3</sup> G <sub>1/2</sub> | 696.1(27)    | 30.00      | F=1: -7815.28                                          | F=2: -6431.61  | F=3: -4352.89  | F=4: -1575.27  | F=5: +1906.40 | F=6: +6098.55 |                |                |
| z <sup>3</sup> G <sub>5/2</sub> | 398.4(18)    | 150.00     | F=1: -4401.87                                          | F=2: -3647.89  | F=3: -2500.84  | F=4: -941.45   | F=5: +1056.01 | F=6: +3523.68 |                |                |
| z <sup>3</sup> D <sub>3/2</sub> | 705(6)       | 30.00      | F=2: -4739.37                                          | F=3: -2647.27  | F=4: +162.20   | F=5: +3706.18  |               |               |                |                |
| z <sup>3</sup> G <sub>7/2</sub> | 495(3)       |            | F=1: -9522.20                                          | F=2: -8532.89  | F=3: -7048.90  | F=4: -5070.27  | F=5: -2596.97 | F=6: +371.00  | F=7: +3833.61  | F=8: +7790.90  |
| z <sup>3</sup> F <sub>5/2</sub> | 448.8(18)    | 180.00     | F=0: -6972.01                                          | F=1: -6541.59  | F=2: -5677.07  | F=3: -4371.11  | F=4: -2612.68 | F=5: -387.10  | F=6: +2324.01  | F=7: +5542.68  |
| y <sup>1</sup> D <sub>5/2</sub> | 478.2(24)    | 30.00      | F=0: -7450.82                                          | F=1: -6987.96  | F=2: -6059.17  | F=3: -4658.33  | F=4: -2776.26 | F=5: -400.72  | F=6: +2483.61  | F=7: +5895.08  |
| y <sup>1</sup> G <sub>5/2</sub> | 300(6)       |            | F=0: -6820.22                                          | F=1: -5920.85  | F=2: -4721.69  | F=3: -3222.74  | F=4: -1424.00 | F=7: +674.53  | F=8: +3072.85  | F=9: +5770.96  |
| y <sup>1</sup> G <sub>3/2</sub> | 334.3(27)    |            | F=1: -6434.70                                          | F=2: -5766.16  | F=3: -4763.35  | F=4: -3426.27  | F=5: -1754.92 | F=6: +250.70  | F=7: +2590.59  | F=8: +5264.75  |
| y <sup>1</sup> D <sub>3/2</sub> | 465(6)       |            | F=1: -5227.65                                          | F=2: -4298.29  | F=3: -2904.25  | F=4: -1045.53  | F=5: +1277.87 | F=6: +4065.95 |                |                |
| y <sup>3</sup> F <sub>7/2</sub> | 297(3)       | 240.00     | F=1: -5603.21                                          | F=2: -5043.91  | F=3: -4197.83  | F=4: -3056.38  | F=5: -1608.15 | F=6: +161.16  | F=7: +2268.69  | F=8: +4734.44  |
| y <sup>3</sup> F <sub>5/2</sub> | 508.1(2)     |            | F=0: -8003.36                                          | F=1: -7495.21  | F=2: -6478.91  | F=3: -4954.46  | F=4: -2921.86 | F=5: -381.11  | F=6: +2667.79  | F=7: +6224.84  |
| y <sup>1</sup> G <sub>5/2</sub> | 441.6(18)    |            | F=1: -8500.61                                          | F=2: -7617.43  | F=3: -6292.66  | F=4: -4526.30  | F=5: -2318.35 | F=6: +331.19  | F=7: +3422.32  | F=8: +6955.04  |
| y <sup>3</sup> G <sub>7/2</sub> | 387(6)       |            | F=0: -6091.00                                          | F=1: -5704.27  | F=2: -4930.81  | F=3: -3770.62  | F=4: -2223.70 | F=5: -290.05  | F=6: +2030.33  | F=7: +4737.44  |
| y <sup>3</sup> G <sub>5/2</sub> | 804.9(27)    | -120.00    | F=1: -9119.86                                          | F=2: -7475.69  | F=3: -5022.30  | F=4: -1775.12  | F=5: +2245.30 | F=6: +7013.23 |                |                |
| (33946) <sub>1/2</sub>          | 603(3)       |            | F=1: -6779.02                                          | F=2: -5573.86  | F=3: -3766.12  | F=4: -1355.81  | F=5: +1657.10 | F=6: +5272.57 |                |                |
| z <sup>1</sup> D <sub>5/2</sub> | 1190.8(24)   | -60.00     | F=2: -8070.04                                          | F=3: -4454.79  | F=4: +325.56   | F=5: +6236.70  |               |               |                |                |
| z <sup>1</sup> D <sub>3/2</sub> | 303.7(21)    |            | F=0: -4783.12                                          | F=1: -4479.43  | F=2: -3872.05  | F=3: -2960.98  | F=4: -1746.22 | F=5: -227.77  | F=6: +1594.37  | F=7: +3720.20  |
| z <sup>1</sup> D <sub>5/2</sub> | 355.2(27)    | 30.00      | F=1: -3980.49                                          | F=2: -3278.56  | F=3: -2222.45  | F=4: -808.31   | F=5: +969.01  | F=6: +3115.94 |                |                |
| z <sup>1</sup> D <sub>3/2</sub> | 528(6)       |            | F=2: -3561.50                                          | F=3: -1978.61  | F=4: +131.91   | F=5: +2770.06  |               |               |                |                |
| z <sup>1</sup> D <sub>5/2</sub> | 1673(9)      |            | F=3: -3763.80                                          | F=4: +2927.40  |                |                |               |               |                |                |
| z <sup>1</sup> F <sub>5/2</sub> | 441(6)       |            | F=1: -8483.28                                          | F=2: -7601.90  | F=3: -6279.83  | F=4: -4517.07  | F=5: -2313.62 | F=6: +330.52  | F=7: +3413.35  | F=8: +6940.87  |
| z <sup>3</sup> G <sub>1/2</sub> | 237(6)       | 240.00     | F=2: -5288.84                                          | F=3: -4617.59  | F=4: -3713.87  | F=5: -2570.19  | F=6: -1177.20 | F=7: +476.32  | F=8: +2403.45  | F=9: +4619.17  |
| z <sup>3</sup> F <sub>5/2</sub> | 531(6)       |            | F=0: -8074.24                                          | F=1: -7561.59  | F=2: -6536.29  | F=3: -4998.34  | F=4: -2947.74 | F=5: -384.49  | F=6: +2691.41  | F=7: +6279.96  |
| z <sup>3</sup> G <sub>3/2</sub> | 330(3)       |            | F=1: -6348.07                                          | F=2: -5688.53  | F=3: -4699.22  | F=4: -3380.14  | F=5: -1731.29 | F=6: +247.33  | F=7: +2555.72  | F=8: +5193.88  |
| z <sup>3</sup> F <sub>3/2</sub> | 660(24)      |            | F=1: -7419.82                                          | F=2: -6100.74  | F=3: -4122.12  | F=4: -1483.96  | F=5: +1813.73 | F=6: +5770.98 |                |                |
| z <sup>1</sup> F <sub>3/2</sub> | 1019(9)      |            | F=2: -6880.27                                          | F=3: -3822.38  | F=4: +254.82   | F=5: +351.32   |               |               |                |                |
| z <sup>3</sup> G <sub>5/2</sub> | 537(27)      |            | F=0: -8451.92                                          | F=1: -7915.29  | F=2: -6842.03  | F=3: -5232.14  | F=4: -3085.62 | F=5: -402.47  | F=6: +2817.31  | F=7: +6573.72  |
| z <sup>3</sup> G <sub>3/2</sub> | 929(9)       |            | F=1: -10455.30                                         | F=2: -8596.58  | F=3: -5808.50  | F=4: -2091.06  | F=5: -2555.74 | F=6: +8131.90 |                |                |
| y <sup>1</sup> D <sub>5/2</sub> | 681(6)       | 60.00      | F=1: -7623.82                                          | F=2: -6279.90  | F=3: -4257.60  | F=4: -1549.19  | F=5: +1855.60 | F=6: +5969.64 |                |                |
| y <sup>1</sup> F <sub>5/2</sub> | 983(6)       | -150.00    | F=1: -18997.66                                         | F=2: -17009.59 | F=3: -14031.95 | F=4: -10070.10 | F=5: -5131.18 | F=6: +775.88  | F=7: +7640.37  | F=8: +15449.79 |
| y <sup>1</sup> D <sub>3/2</sub> | 711(6)       |            | F=0: -11190.53                                         | F=1: -10480.02 | F=2: -9059.00  | F=3: -6927.47  | F=4: -4085.43 | F=5: -532.88  | F=6: +7370.18  | F=7: +8703.75  |
| z <sup>1</sup> F <sub>3/2</sub> | 711(9)       | -150.00    | F=0: -11270.89                                         | F=1: -10545.07 | F=2: -9096.50  | F=3: -6931.30  | F=4: -4058.65 | F=5: -490.79  | F=6: +7356.96  | F=7: +8666.25  |
| (43848) <sub>1/2</sub>          | 779(3)       |            | F=0: -12276.50                                         | F=1: -11497.03 | F=2: -9938.11  | F=3: -7599.73  | F=4: -4481.90 | F=5: -584.60  | F=6: +4092.16  | F=7: +9548.39  |
| z <sup>1</sup> G <sub>1/2</sub> | 982.1(27)    | -150.00    | F=2: -22405.28                                         | F=3: -19434.37 | F=4: -15478.62 | F=5: -10542.69 | F=6: -4632.44 | F=7: +2225.13 | F=8: +10081.83 | F=9: +18688.31 |
| (44202) <sub>1/2</sub>          | 660(24)      |            | F=1: -7419.82                                          | F=2: -6100.74  | F=3: -4122.12  | F=4: -1483.96  | F=5: +1813.73 | F=6: +5770.98 |                |                |
| z <sup>1</sup> G <sub>5/2</sub> | 546(12)      |            | F=0: -8593.51                                          | F=1: -8047.90  | F=2: -6956.65  | F=3: -5319.80  | F=4: -3137.32 | F=5: -409.21  | F=6: +2864.51  | F=7: +6683.85  |
| z <sup>1</sup> F <sub>3/2</sub> | 435(9)       | 180.00     | F=2: -2837.80                                          | F=3: -1662.27  | F=4: +25.10    | F=5: +327.18   |               |               |                |                |

Measured A, B constants; sources in the data file.